Evgenii Nikishin

Montreal, QC, Canada
nikishin.evg@gmail.com
evgenii-nikishin.github.io
Google Scholar

Education

2021–Now Ph.D. student in Computer Science.
Mila, University of Montreal
Advisors: Pierre-Luc Bacon, Aaron Courville

2019–2020 Ph.D. student in Operations Research.
Cornell University, transferred to Mila

2017–2019 M.Sc. in Computer Science, with honors (GPA: 8.9/10.0).
Higher School of Economics (HSE) and
Skolkovo Institute of Science and Technology
Advisor: Dmitry Vetrov

2013–2017 B.Sc. in Computer Science, with honors (GPA: 4.9/5.0).
Lomonosov Moscow State University
Advisor: Alexander D'yakonov

Experience

May 2020–Aug 2020 Research Intern (remote), **Mila, University of Montreal**.
Addressing the objective mismatch in model-based RL. Hosts: Pierre-Luc Bacon, Yoshua Bengio

Sep 2017–Aug 2019 Research Assistant, **Bayesian Methods Research Group**.
A part of **Samsung AI Lab**
Improving stability and transfer in deep RL. Supervisor: Dmitry Vetrov

Jul 2018–Aug 2018 Research Intern, **ETH Zürich**.
Summer Research Fellowship (a total of 16 fellows among 1k applicants). Host: Gunnar Rätsch

Jul 2016–Dec 2016 Research & Development Intern, **Kaspersky Lab**.
Improved feature extraction from textual representations of files for malicious software detection

Selected awards

2021 DIRO excellence scholarship.
2021 ICLR outstanding reviewer.
2020 Mitacs-Mila short-term research grant.
2019 Cornell graduate fellowship.
2019 NeurIPS 2019 travel award.
2019 MLSS 2019 full travel grant.
2016 Russian state academic excellence scholarship.
2016 Second place in Yandex Chatbot Hackathon.
2015 Bronze medal in Driver Telematics Analysis competition on Kaggle.

Publications

2022 The Primacy Bias in Deep Reinforcement Learning .
Evgenii Nikishin*, Max Schwarzer*, Pierluca D'Oro*, Pierre-Luc Bacon, Aaron Courville
We identify a damaging tendency of deep RL agents to overfit early experiences and propose a simple yet powerful remedy based on periodic resetting of a part of the agent
ICML 2022; also RLDM 2022. Code

2021 Control-Oriented Model-Based Reinforcement Learning with Implicit Differentiation.
Evgenii Nikishin, Romina Abachi, Rishabh Agarwal, Pierre-Luc Bacon
A model learning method for RL that directly optimizes the sum of rewards instead of likelihood, a proxy to the agent's objective
AAAI 2022; also in Unsupervised RL Workshop, ICML 2021. Code

2019 Unsupervised Domain Adaptation with Shared Latent Dynamics for Reinforcement Learning.
Evgenii Nikishin, Arsenii Ashukha, Dmitry Vetrov
Domain adaptation via learning shared dynamics in a latent space and adversarial matching of the latent representations of states
Bayesian Deep Learning workshop, NeurIPS 2019. Code

2018 Improving Stability in Deep Reinforcement Learning with Weight Averaging.
Evgenii Nikishin, Pavel Izmailov, Ben Athiwaratkun, Dmitrii Podoprikhin,
Timur Garipov, Pavel Shvechikov, Dmitry Vetrov, Andrew Gordon Wilson
Averaging weights during training to reduce the variance of policy gradient estimates
Uncertainty in Deep Learning workshop, UAI 2018

Intived talks

Jun 2022 **Microsoft Research**. The Primacy Bias in Deep Reinforcement Learning.
May 2022 **ML Collective**. The Primacy Bias in Deep Reinforcement Learning.
Mar 2022 **Cornell Tech**. Control-Oriented MBRL with Implicit Differentiation.
Dec 2021 **AI Research Institute Moscow**. Control-Oriented MBRL with Implicit Differentiation.
Sep 2021 **Amsterdam ML Lab**. Control-Oriented MBRL with Implicit Differentiation.
Aug 2021 **Stanford University**. Control-Oriented MBRL with Implicit Differentiation.
Jul 2021 **ML Collective**. Control-Oriented MBRL with Implicit Differentiation.
Jul 2021 **Google Brain**. Control-Oriented MBRL with Implicit Differentiation.
Jun 2021 **University of Bristol**. Control-Oriented MBRL with Implicit Differentiation.

Other research projects

2018 **Reinforcement Learning for Personal Treatment**.
Off-policy policy gradient methods for treatment of patients in intensive care units

2017 **Hierarchical Imitation Learning**.
Generative Adversarial Imitation Learning for option-augmented policies

2017 **Stable Matrix Approximation via Autoencoders**.
Autoencoder-based architecture for recommender systems with stability guarantees. Code

2016 **Community Detection Methods in Social Graphs**.
Review and comparison of overlapping and non-overlapping community detection algorithms
Accepted to Lomonosov 2017 conference. Code

Service

Organizing: **Decision Awareness in RL** workshop at ICML 2022, **DeepBayes** summer school 2018, 2019.

Reviewer: ICML 2022, ICLR 2022, NeurIPS 2021 workshop proposals, NeurIPS 2021, ICML 2021, ICLR 2021, AAAI 2021, ICML 2020 BIG workshop, IndabaX-AI4D innovation grants 2020, ICLR 2020.

Volunteer: ICLR 2020, NeurIPS 2019 WiML workshop.

Admissions: Cornell CS PhD applicants starting in Fall 2020.

Teaching

Sep 2020–Dec 2020 Teaching Assistant, **Cornell University**.
ORIE 4350, Introduction to Game Theory

Sep 2017–Dec 2017 Practice Lecturer, **Higher School of Economics**.
Machine Learning classes for 3rd year CS students

Jul 2017 Teaching Assistant, **Coursera**.
Quizzes and assignments for Bayesian Methods for Machine Learning course.
The specialization of the course received Coursera Outstanding Educator Award

Additional education

Aug 2020 (Virtual) DL and RL Summer School, Mila, Canada.
Jul 2019 Machine Learning Summer School, Gatsby Unit, UK.
Jul 2019 Reinforcement Learning Summer School, INRIA, France.
Mar 2017 Spring school "Structural Inference", Humboldt University, Germany.
Feb 2017 Winter school on Reinforcement Learning, MIPT, Russia.